

ROS2 - CLI midlevel

Intelligent Robotics I

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Parameters

- configuration value of a node
- each node maintains its own parameters
- node stores parameters as integers, floats, boolans, strings and lists

```
# ----- terminal 1 ----- #  
ros2 run turtlesim turtlesim_node  
  
# ----- terminal 2 ----- #  
ros2 run turtlesim turtle_teleop_key
```

```
# ----- terminal 3 ----- #
```

```
ros2 param list
```

```
# -----
```

```
/teleop_turtle:
```

```
  qos_overrides./parameter_events.publisher.depth
```

```
  qos_overrides./parameter_events.publisher.durability
```

```
  scale_angular
```

```
  scale_linear
```

```
  use_sim_time
```

```
/turtlesim:
```

```
  background_b
```

```
  background_g
```

```
  background_r
```

```
  qos_overrides./parameter_events.publisher.depth
```

```
  qos_overrides./parameter_events.publisher.durability
```

```
  use_sim_time
```

Displays the type and current value of a parameter.

```
# ros2 param get <node_name> <parameter_name>
```

```
# ----- terminal 3 ----- #
```

```
ros2 param get /turtlesim background_g
```

```
# -----
```

```
Integer value is: 86
```

Changes parameters value at runtime.

```
# ros param set <node_name> <parameter_name> <value>
```

```
# ----- terminal 3 ----- #
```

```
ros2 param set /turtlesim background_r 150
```

```
# -----
```

```
Set parameter successful
```


Views all parameters values of the node.

```
# ros2 param dump <node_name>

# ----- terminal 3 ----- #
# saves the dump into yaml file
ros2 param dump /turtlesim > turtlesim.yaml
```

Loads parameters from a file into running node.

```
# ros2 param load <node_name> <parameter_file>
```

```
# ----- terminal 3 ----- #
```

```
# saves the dump into yaml file
```

```
ros2 param load /turtlesim turtle.yaml
```

```
# ros2 run <package_name> <executable_name> \  
# --ros-args --params-file turtlesim.yaml
```

```
# ----- terminal 1 ----- #  
# stop turtlesim  
ros2 run turtlesim turtlesim_node \  
--ros-args --params-file turtlesim.yaml
```

Launch



- launch system sets up and runs your robot system
- defines what nodes to run and their arguments
- launch files can be written in Python, XML, or YAML
- it will be only seen the YAML launch files

what is yaml?

- YAML stands for “Yet Another Markup Language”
- popular data serialization language
- often used for configuration files

```
robot:
  name: "ExplorerBot"
  type: "Mobile Robot"
  parts:
    sensors:
      - name: "Lidar"
        type: "Laser Scanner"
        range: "10m"
    actuators:
      - name: "Wheels"
        type: "Motorized"
        count: 4
    controller:
      type: "Microcontroller"
      model: "Raspberry Pi 4"
      OS: "ROS 2 Humble"
```

Paste this code into `launch/turtlesim_mimic_launch.yaml`

launch:

launch one turtlesim window

- node:

```
pkg: "turtlesim"
exec: "turtlesim_node"
name: "sim"
namespace: "turtlesim1"
```

launch other turtlesim window

- node:

```
pkg: "turtlesim"
exec: "turtlesim_node"
name: "sim"
namespace: "turtlesim2"
```

- node:

```
pkg: "turtlesim"
exec: "mimic"
name: "mimic"
remap:
-
  from: "/input/pose"
  to: "/turtlesim1/turtle1/pose"
-
  from: "/output/cmd_vel"
  to: "/turtlesim2/turtle1/cmd_vel"
```

```
# ros2 launch <yaml_file>
```

```
cd launch
```

```
ros2 launch turtlesim_mimic_launch.yaml
```

```
# two turtlesim windows will open:
```

```
[INFO] [launch]: Default logging verbosity is set to INFO
```

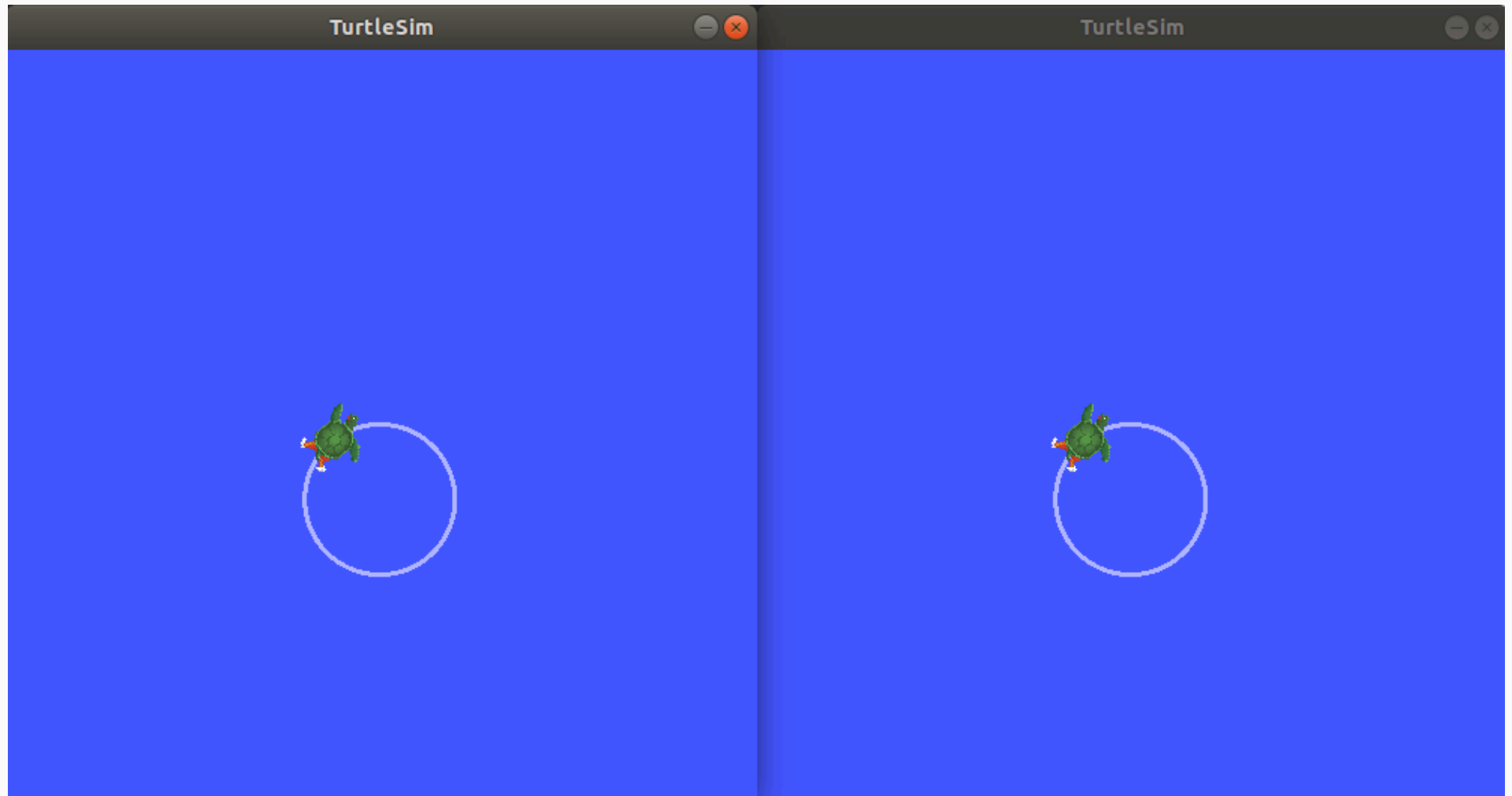
```
[INFO] [turtlesim_node-1]: process started with pid [11714]
```

```
[INFO] [turtlesim_node-2]: process started with pid [11715]
```

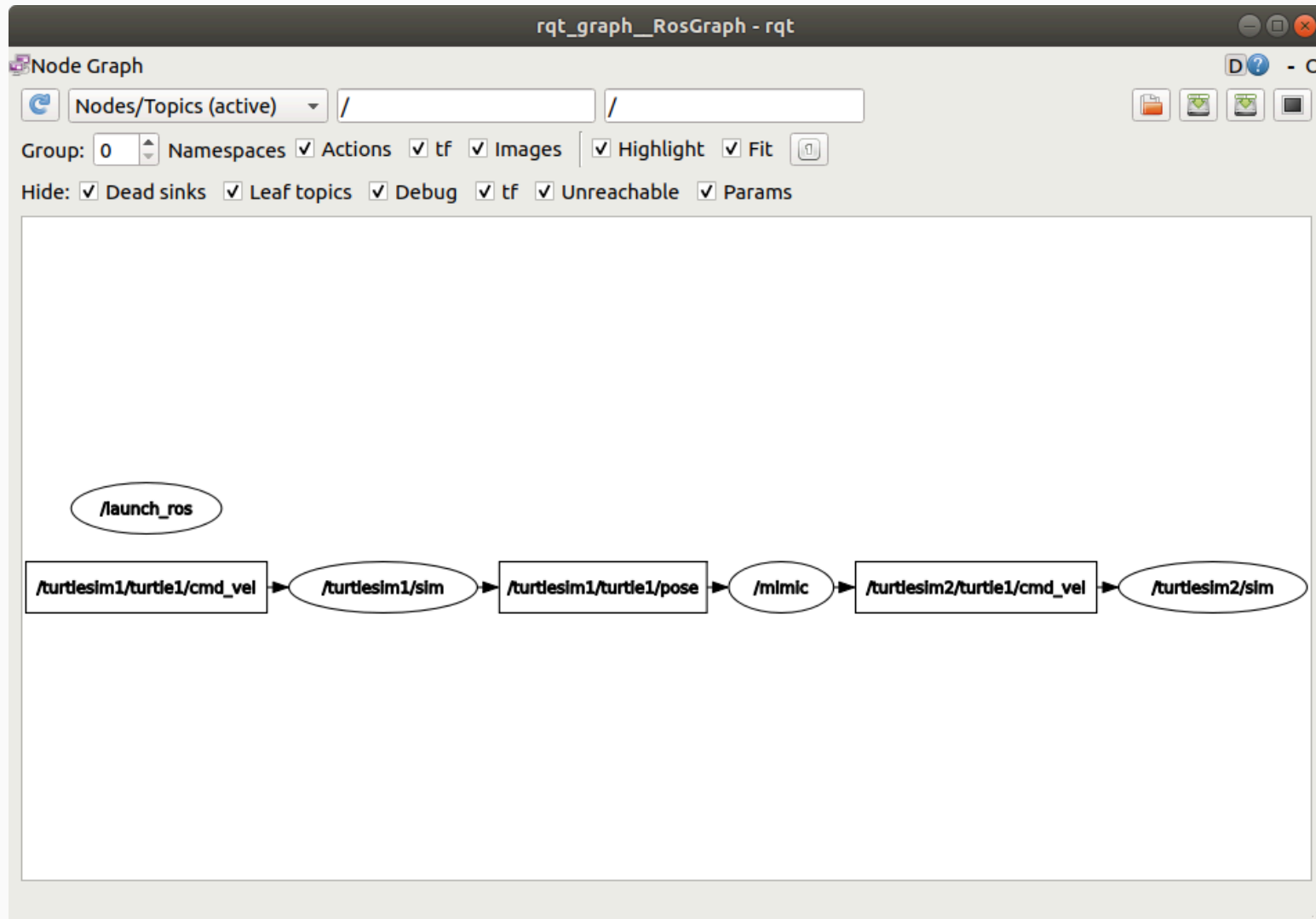
```
[INFO] [mimic-3]: process started with pid [11716]
```


launch 2/

```
ros2 topic pub -r 1 \  
/turtlesim1/turtle1/cmd_vel geometry_msgs/msg/Twist \  
"{linear: {x: 2.0, y: 0.0, z: 0.0}, \  
angular: {x: 0.0, y: 0.0, z: -1.8}}"
```



rqt_graph



Bag



- record data published on topics
- the data can be replayed
- records help to analyze and share

```
# ----- terminal 1 ----- #  
ros2 run turtlesim turtlesim_node  
  
# ----- terminal 2 ----- #  
ros2 run turtlesim turtle_teleop_key
```

```
# ----- terminal 3 ----- #  
ros2 topic echo /turtle1/cmd_vel  
# -----  
# active the terminal where is teleop  
# and use the arrow keys  
linear:  
  x: 2.0  
  y: 0.0  
  z: 0.0  
angular:  
  x: 0.0  
  y: 0.0  
  z: 0.0  
---
```

- press Ctrl+C to stop recording.
- data will be accumulated in “bag” directory

```
# ros2 bag record <topic_name>
```

```
# ----- terminal 4 ----- #
```

```
ros2 bag record /turtle1/cmd_vel
```

```
# -----
```

```
# `ros2 bag` is recording data published on `/turtle1/cmd_vel`
```

```
# return to teleop terminal and move the turtle
```

```
[INFO] [rosbag2_storage]: Opened database 'rosbag2_2019_10_11-05_18_45'.
```

```
[INFO] [rosbag2_transport]: Listening for topics
```

```
[INFO] [rosbag2_transport]: Subscribed to topic '/turtle1/cmd_vel'
```

```
[INFO] [rosbag2_transport]: All requested topics are subscribed...
```

record multiple topics

```
# ros2 bag record -o <file> <topic_1> ... <topic_n>

# ----- terminal 4 ----- #
ros2 bag record -o subset /turtle1/cmd_vel /turtle1/pose
# -----
[INFO] [rosbag2_storage]: Opened database 'subset'.
[INFO] [rosbag2_transport]: Listening for topics...
[INFO] [rosbag2_transport]: Subscribed to topic '/turtle1/cmd_vel'
[INFO] [rosbag2_transport]: Subscribed to topic '/turtle1/pose'
[INFO] [rosbag2_transport]: All requested topics are subscribed.
```



```
# ros2 bag info <bag_file_name>
```

```
# ----- terminal 4 ----- #
```

```
ros2 bag info subset
```

```
# -----
```

```
Files:                subset.db3
```

```
Bag size:             228.5 KiB
```

```
Storage id:           sqlite3
```

```
Duration:             48.47s
```

```
Start:                Oct 11 2019 06:09:09.12 (1570799349.12)
```

```
End                   Oct 11 2019 06:09:57.60 (1570799397.60)
```

```
Messages:             3013
```

```
Topic information: ...
```

- stop the teleop by pressing Ctrl+C
- make sure the turtlesim window is visible
- the command runs as long as turtlesim is active

```
# ----- terminal 3 -----
```

```
ros2 bag play subset #"subset" is the bag filename
```

```
# -----
```

```
[INFO] [rosbag2_storage]: Opened database 'subset'.
```

RQT Console

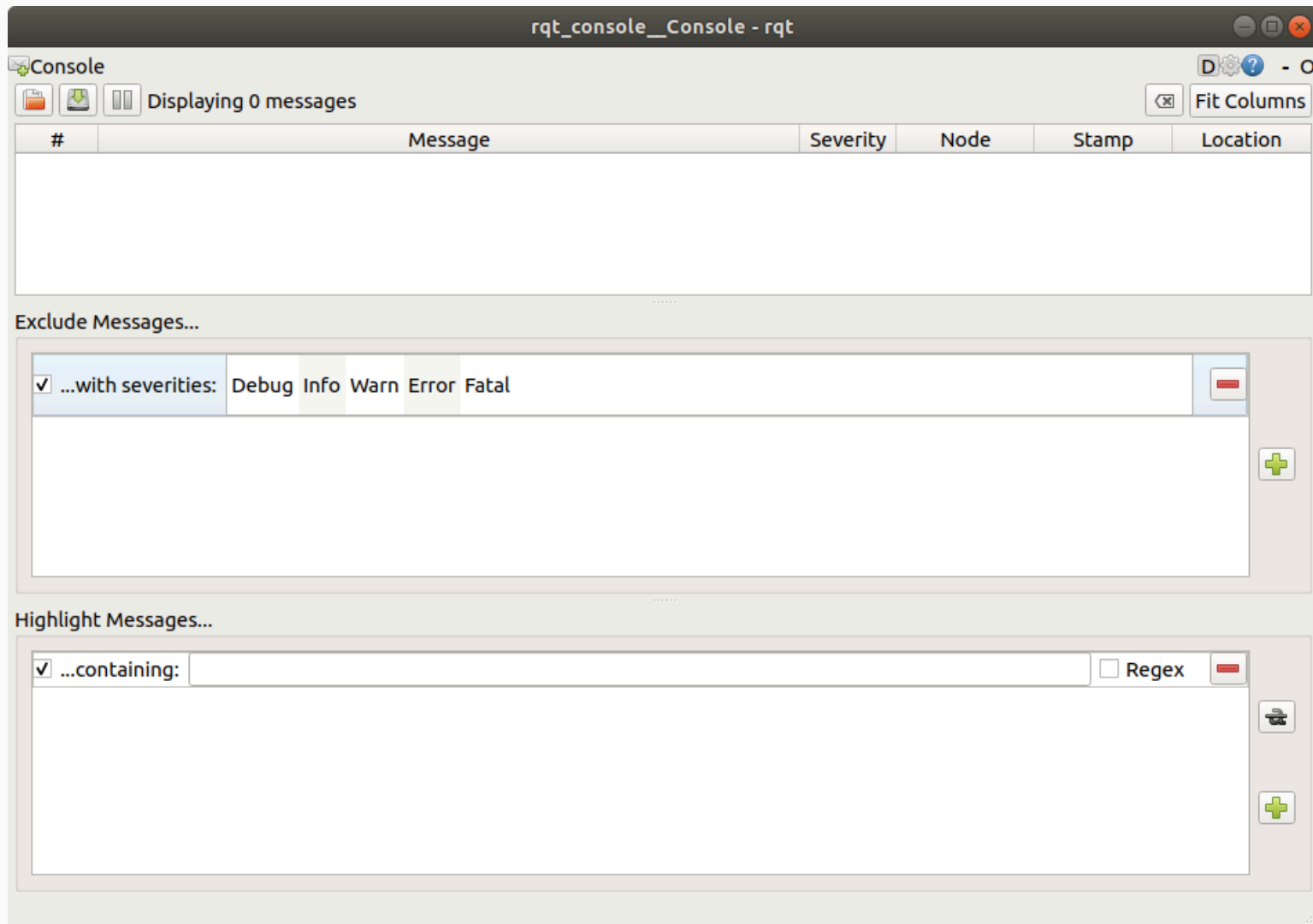


what?

- `rqt_console` collect and organize log messages
- allows to filter, save and reload them

setup

```
# ----- terminal 1 ----- #  
ros2 run rqt_console rqt_console
```



messages

```
# ----- terminal 2 ----- #  
ros2 topic pub -r 1 /turtle1/cmd_vel  
geometry_msgs/msg/Twist \  
"{linear: {x: 2.0, y: 0.0, z: 0.0}, \  
angular: {x: 0.0,y: 0.0,z: 0.0}}"
```

The screenshot shows the `rqt_console__Console - rqt` window. The title bar indicates it is displaying 342 messages. The main table lists messages with columns for #, Message, Severity, Node, Stamp, and Location. The messages shown are warnings from the `turtlesim` node, stating "Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])". Below the table, there are sections for "Exclude Messages..." and "Highlight Messages...". The "Exclude Messages..." section has a checkbox for "...with severities:" which is checked, and a dropdown menu showing "Debug", "Info", "Warn", "Error", and "Fatal". The "Highlight Messages..." section has a checkbox for "...containing:" which is checked, and a text input field. There is also a "Regex" checkbox.

#	Message	Severity	Node	Stamp	Location
#342	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtlesim	16:28:39.212...	/home/miche...
#341	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtlesim	16:28:39.197...	/home/miche...
#340	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtlesim	16:28:39.180...	/home/miche...
#339	Oh no! I hit the wall! (Clamping from [x=11.104889, y=5.544445])	Warn	turtlesim	16:28:39.164...	/home/miche...

References

- [Ros Documentation Humble - Beginner CLI tools](<https://docs.ros.org/en/humble/Tutorials.html>)